

Mercer General ED Digital Triage Model

Introduction

The purpose of this project is to create a digital triage support system for the Emergency Department (ED) at the Mercer General Hospital. This system will assist the triage nurse by analysing patient information that was collected during intake and assigning an Emergency Severity Index (ESI) category from 1-5.

Operational Constraints & System Boundaries

- **The Mercer Assumption (Nurse-Led triage):** This system explicitly assumes operation by a trained triage nurse during initial front-door assessment. It is optimised to supplement clinical intuition, not replace it.
- **The 3-to-5 Clinician Attention Budget:** At the intake desk, data points are constrained; the tool operates *exclusively* on data available during this brief window: Chief Complaint (CC), vital signs, brief historical context, and a pain score. It does not factor in laboratory or imaging results, as these represent downstream operational bottlenecks.
- **The Re-Triage Mandate:** Acuity is dynamic. This tool implements a time-decayed background monitoring loop to ensure patients waiting in the fast-track or acute queues are flagged for physical re-assessment, preventing digital blind spots.

Understanding the Mercer ED Workflow

**ARRIVAL → REGISTRATION → TRIAGE → ZONE → ASSESSMENT → DISPOSITION
→ EXIT**

The proposed AI tool operates during the TRIAGE stage.

At this point, the nurse already knows:

- Demographics,
- Chief complaint,
- Allergies,
- Brief history,
- And initial vital signs.

Therefore, the tool must:

- Work quickly,
- Avoid depending on lab tests,
- And provide a recommendation within the nurse's 3-5 minute assessment window.

Clinical Importance of Each Vital Sign

Vital Sign	Normal Range	Clinical relevance	Danger Zone
Glasgow Coma Scale (GSC)	3- 15 score	Measures level of consciousness and neurological function. Helps detect brain injury, stroke, overdose, or severe illness.	- Below 8
Systolic Blood Pressure (SBP)	50-250 mmHg	Indicates pressure during heart contraction. Helps identify shock, bleeding, hypertension, or cardiac problems.	- Above 250 mmHg - Below 50 mmHg
Diastolic Blood Pressure (DBP)	30-150 mmHg	Measures pressure when the heart relaxes. Important for organ perfusion and cardiovascular stability.	- Above 150 mmHg - Below 30 mmHg
Mean Arterial Pressure (MAP)	40-180 mmHg	Indicates overall blood flow and organ perfusion. Frequently used in critical care.	- Above 180 mmHg - Below 40 mmHg
Pulse (Heart Rate)	60-100 bpm	Reflects cardiac activity. High pulse may indicate pain, infection, dehydration or shock. Low pulse may indicate cardiac dysfunction.	- Above 250 bpm - Below 20 bpm
Temperature	32.0 °C - 43.0 °C	Detects fever, infection, sepsis,	- Above 43.0 °C

		hypothermia or inflammatory conditions.	- Below 32.0 °C
Respiratory Rate (RR)	5-60 breaths/minute	One of the earliest indicators of deterioration. High RR suggests respiratory distress or sepsis.	- Above 60 b/p - Below 5 b/p
Fraction of Inspired Oxygen (FiO ₂)	21-100%	Measures oxygen support required by the patient. Higher FiO ₂ means greater respiratory support is needed.	- Below 21%

ESI Classification Logic

Mercer uses the ESI 5-level system.

ESI Level	Meaning	Target Time
1	Life-threatening	Immediate
2	High-risk / danger vitals	≤10 min
3	Stable but needs multiple resources	≤ 30 min
4	Minor condition needing one resource	≤ 60 min
5	Very minor complaint	Can wait

This algorithm mirrors this structure.

Pseudocode Algorithm

START

INPUT:

Age

Gender

GCS

SBP

DBP

MAP

pulse

Temp

RR

Fio2

STEP 1 — VALIDATE INPUT RANGES

IF Age < 0 OR Age > 120

 DISPLAY "Invalid Age"

 STOP

STEP 2 — CHECK FOR CRITICAL (ESI-1)

IF GCS <= 8

 OR SBP < 90

 OR MAP < 65

 OR pulse > 140

 OR RR > 35

 OR Temp > 40

 OR Fio2 >= 60

THEN

 ESI = 1

STEP 3 — CHECK FOR HIGH RISK (ESI-2)

ELSE IF GCS BETWEEN 9 AND 12
OR SBP BETWEEN 90 AND 100
OR MAP BETWEEN 65 AND 70
OR pulse BETWEEN 120 AND 140
OR RR BETWEEN 25 AND 35
OR Temp BETWEEN 39 AND 40
OR Fio2 BETWEEN 40 AND 59

THEN

ESI = 2

STEP 4 — CHECK FOR URGENT (ESI-3)

ELSE IF pulse BETWEEN 100 AND 119
OR RR BETWEEN 21 AND 24
OR Temp BETWEEN 37.5 AND 38.9
OR mild abnormal vitals present

THEN

ESI = 3

STEP 5 — CHECK FOR SEMI-URGENT (ESI-4)

ELSE IF stable vitals
AND requires one clinical resource

THEN

ESI = 4

STEP 6 — NON-URGENT (ESI-5)

ELSE

ESI = 5

STEP 7 — RE-TRIAGE LOGIC

IF waiting patient shows worsening vitals

THEN

repeat assessment

update ESI level

STEP 8 — OUTPUT RESULT

DISPLAY ESI level

DISPLAY triage recommendation

END

Logical Flow of the System

The algorithm prioritizes the sickest patients first.

ESI 1 – Immediate Threat

Patients with severe oxygen loss, shock, cardiac arrest, or extreme vitals are classified immediately as ESI 1.

These patients go directly to the resuscitation zone.

ESI 2 – High Risk

Patients who are not yet crashing but show danger signs are classified as ESI 2.

Examples:

- chest pain,
- suspected stroke,
- severe pain,
- dangerously abnormal vitals.

These patients must be seen within 10 minutes.

ESI 3 – Urgent

These patients are stable but likely require multiple resources such as:

- blood tests,
- Imaging,
- IV medications,
- monitoring.

ESI 4–5 – Lower Acuity

Minor complaints with limited resource needs are categorized lower priority and directed to fast-track care.

Why Re-Triage Is Essential

Mercer specifically states that re-triage is part of current practice.

This means:

- a patient may initially appear stable,
- but deteriorate while waiting.

For example:

- HR rises sharply,
- or new confusion develops.

The AI system therefore includes automatic reassessment reminders and allows the nurse to update the ESI category.

Without re-triage support, the digital system would perform worse than Mercer's current workflow.

Design Decisions and Rationale

1. Fast Decision-Making

The system avoids lab-dependent logic because laboratory results are unavailable during triage.

This follows Mercer's workflow constraints.

2. Simple IF/ELSE Structure

The logic is transparent and explainable.

In healthcare, clinicians must understand:

- why a decision was made,
- and what triggered escalation.

3. Focus on Vital Signs

Vital signs are universally available at triage and provide rapid evidence of deterioration.

This makes them ideal for an AI-supported frontline screening tool.

4. Human Oversight

The tool supports — not replaces — the triage nurse.

The final decision still belongs to the clinician.

Clinical Benefits of the System

Potential advantages include:

- Faster identification of critically ill patients
- Reduced waiting-room deterioration
- More consistent ESI assignments
- Support for overwhelmed triage staff
- Improved patient safety

The system also aligns with Mercer's operational structure and time constraints.

Use of AI Tools

AI tools were used to assist with:

- organizing the workflow structure,
- refining pseudocode formatting,
- and improving clarity of explanation.

However, all clinical reasoning, threshold interpretation, and workflow decisions were reviewed and understood before inclusion. The final model reflects independent understanding of ED triage operations and emergency prioritisation.

Conclusion

This digital triage model demonstrates how patient vital signs and presenting symptoms can be processed quickly to support ESI classification in the Mercer Emergency Department.

The design reflects:

- real ED workflow constraints,
- nurse-led triage practice,
- emergency clinical reasoning,
- and the importance of continuous reassessment.

By integrating vital signs, risk thresholds, and re-triage logic, the system can improve patient prioritization while supporting clinicians in delivering timely emergency care.

Use of Artificial Intelligence Tools

Generative AI tools were used during the development of this project to assist with:

- Organising workflow structure,
- Refining pseudocode formatting,
- Improving readability,
- And explaining clinical concepts in plain language.

All outputs were critically reviewed, modified where necessary and integrated based on independent understanding of the subject matter.